Comparing models with ANOVA (answer key)

Each part has two scenarios with the same tasks. We'll work through scenario **A** together as a class. Then you'll work through scenario **B** in your group.

Every question in this handout compares a **simpler** model to a more **complex** model. Refer back to this box as you go.

$$F = \frac{(\text{RSS}_{simple} - \text{RSS}_{complex}) \ / \ (p_{complex} - p_{simple})}{\text{RSS}_{complex} \ / \ (n - p_{complex})}$$

- RSS: residual sum of squares, $\sum(y_i - \hat{y}_i)^2$. The variation the model leaves unexplained.
- p: number of coefficients the model estimates, including the intercept
- n: number of observations
- $df = n - p$: degrees of freedom left over after fitting the model
- **Numerator**: improvement in fit per df spent. **Denominator**: leftover noise per df remaining.

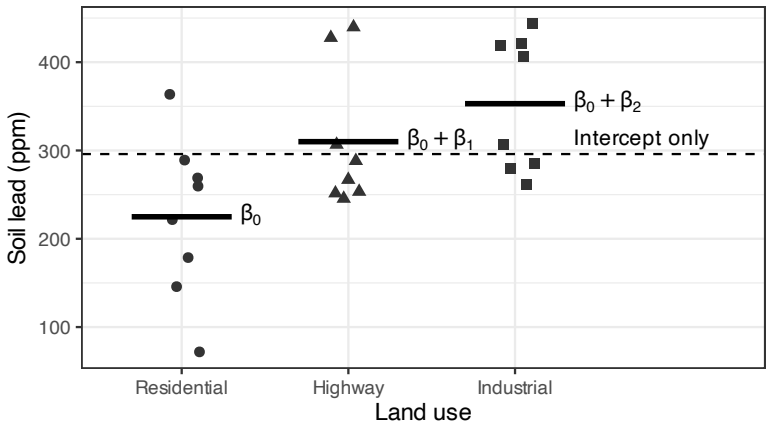
Part 1: Three levels

For each scenario:

1. **Dummy variables.** Fill in the dummy variable table.
2. **Full model.** Write the equation for the full model.
3. **Draw the means.** Use the coefficients to draw the intercept-only model and each category's mean on the figure.
4. **Degrees of freedom and F.** Fill in p and df for each model, then calculate F.
5. **Interpretation.** State H_0 and H_A. Then interpret the p-value for your F in a sentence.

1A. Soil lead (together)

Lead from old gasoline and industrial emissions settles into soil, where children can be exposed. A county health department sampled soil **lead** (ppm) in 8 yards near each of three land uses (**land_use**): residential streets, a highway, and an industrial site.



Dummy variables

Land use	Highway	Industrial
Residential	0	0
Highway	1	0
Industrial	0	1

Full model

$$\text{lead} = \beta_0 + \beta_1 (\text{land use = Highway}) + \beta_2 (\text{land use = Industrial}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

Draw the means

```
coef(lm(lead ~ 1, yards))
```

```
(Intercept)
   295.9625
```

```
coef(lm(lead ~ land_use, yards))
```

```
      (Intercept)    land_useHighway land_useIndustrial
         224.9125            85.0625           128.0875
```

See figure above

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`lead ~ 1`	1	23	210,538.5
`lead ~ land_use`	3	21	142,556.7

$$F = \frac{(210,538.5 - 142,556.7) \ / \ (3-1)}{142,556.7 \ / \ (24-3)} = 5.01$$

Interpretation

State H_0 and H_A.

H_0: $\beta_1 = \beta_2 = 0$

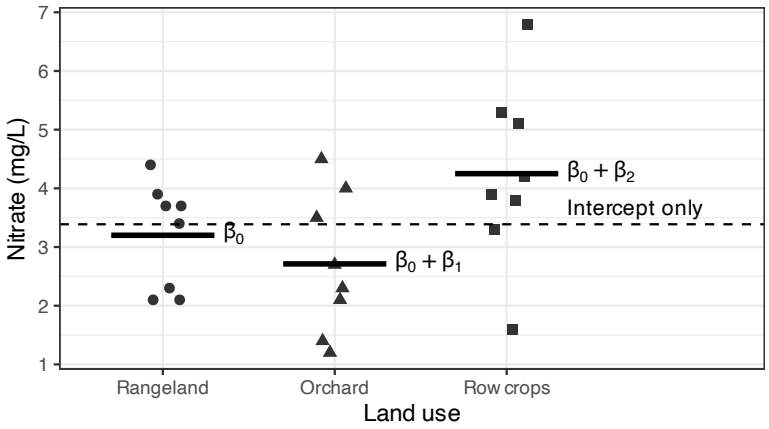
H_A: at least one β is different than 0.

The p-value for this F is p = 0.017. Interpret it in a sentence.

If land use had no relationship with soil lead, we'd see an F at least this large with probability 0.017, so we reject H_0 and conclude there is evidence for soil lead varying by land use.

1B. Groundwater nitrate (in groups)

Fertilizer applied to crops can leach into groundwater as nitrate, contaminating drinking water wells. A water agency measured **nitrate** (mg/L) in 8 wells under each of three land uses (**land_use**): rangeland, orchards, and row crops.



Dummy variables

Land use	Orchard	Row crops
Rangeland	0	0
Orchard	1	0
Row crops	0	1

Full model

$$\text{nitrate} = \beta_0 + \beta_1 (\text{land use = Orchard}) + \beta_2 (\text{land use = Row crops}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

Draw the means

```
coef(lm(nitrate ~ 1, wells))
```

```
(Intercept)
     3.3875
```

```
coef(lm(nitrate ~ land_use, wells))
```

```
      (Intercept)   land_useOrchard land_useRow crops
           3.2000           -0.4875            1.0500
```

See figure above

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`nitrate ~ 1`	1	23	42.2
`nitrate ~ land_use`	3	21	32.3

$$F = \frac{(42.2 - 32.3) \ / \ (3 - 1)}{32.3 \ / \ (24 - 3)} = 3.21$$

Interpretation

State H_0 and H_A.

H_0: $\beta_1 = \beta_2 = 0$

H_A: at least one β is different than 0.

The p-value for this F is p = 0.061. Interpret it in a sentence.

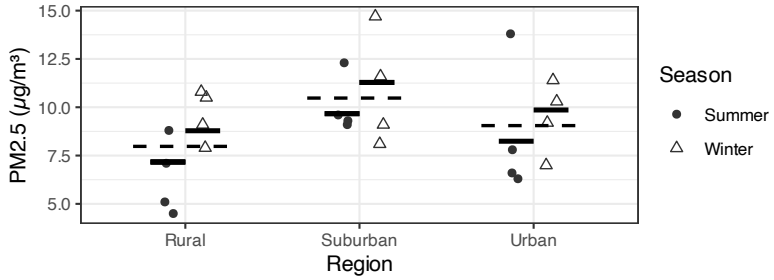
If land use had no relationship with nitrate, we'd see an F at least this large with probability 0.061, so we fail to reject H_0. There is not enough evidence to conclude that nitrate varies by land use.

Part 2: Two categorical predictors

Now there are two categorical predictors. Does the second one justify its cost *after* accounting for the first? Same tasks as Part 1, minus the dummy table.

2A. Air quality (together)

Fine particulate matter (PM2.5) damages lungs and hearts, and winter wood smoke can trap it near the ground. An air district measured PM2.5 (**pm**, µg/m³) on 4 days for each combination of **region** (rural, suburban, or urban) and **season** (summer or winter). Does season matter after accounting for region?



Full model

$$\text{pm} = \beta_0 + \beta_1\,(\text{region} = \text{Suburban}) + \beta_2\,(\text{region} = \text{Urban}) + \beta_3\,(\text{season} = \text{Winter}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

Draw the models

```
coef(lm(pm ~ region, air))
```

```
   (Intercept) regionSuburban    regionUrban
         7.975          2.500          1.075
```

```
coef(lm(pm ~ region + season, air))
```

```
   (Intercept) regionSuburban    regionUrban   seasonWinter
      7.166667       2.500000       1.075000       1.616667
```

See figure above

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`pm ~ region`	3	21	120.4
`pm ~ region + season`	4	20	104.7

Adding season costs 1 df.

$$F = \frac{(120.4 - 104.7)\ /\ (4 - 3)}{104.7\ /\ (24 - 4)} = 2.99$$

Interpretation

State H_0 and H_A.

H_0: $\beta_3 = 0$

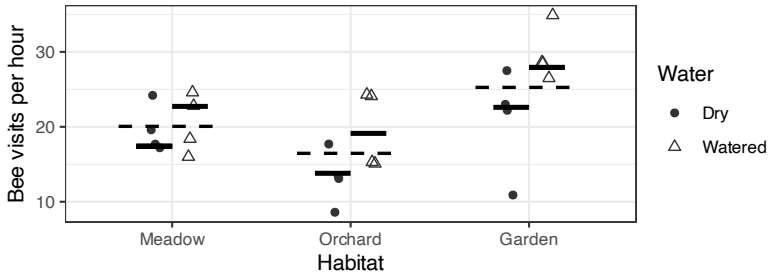
H_A: β_3 is different than 0.

The p-value for this F is p = 0.099. Interpret it in a sentence.

If season had no relationship with PM2.5 within each region, we'd see an F at least this large with probability 0.099, so we fail to reject H_0. There is not enough evidence to conclude that PM2.5 varies by season after accounting for region.

2B. Bees and watering (in groups)

Watered flowers may produce more nectar, attracting more bees. A pollinator team counted bee **visits** per hour to 4 flower patches for each combination of **habitat** (meadow, orchard, or garden) and **water** (dry or watered). Does watering matter after accounting for habitat?



Full model

$$\text{visits} = \beta_0 + \beta_1\,(\text{habitat = Orchard}) + \beta_2\,(\text{habitat = Garden}) + \beta_3\,(\text{water = Watered}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

Draw the models

```
coef(lm(visits ~ habitat, bees))
```

```
   (Intercept) habitatOrchard  habitatGarden
       20.0625        -3.6000         5.2000
```

```
coef(lm(visits ~ habitat + water, bees))
```

```
   (Intercept) habitatOrchard  habitatGarden   waterWatered
        17.400         -3.600          5.200          5.325
```

See figure above

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`visits ~ habitat`	3	21	626.7
`visits ~ habitat + water`	4	20	456.5

Adding watering costs 1 df.

$$F = \frac{(626.7 - 456.5)\ /\ (4-3)}{456.5\ /\ (24-4)} = 7.45$$

Interpretation

State H_0 and H_A.

H_0: $\beta_3 = 0$

H_A: β_3 is different than 0.

The p-value for this F is p = 0.013. Interpret it in a sentence.

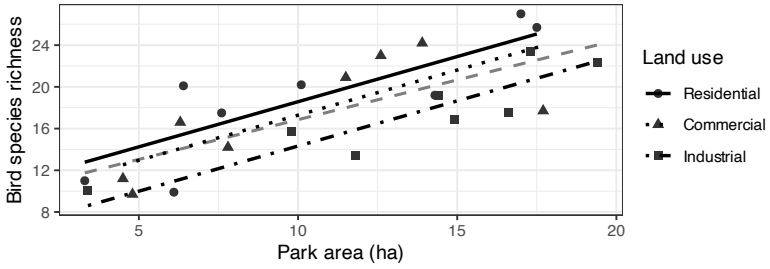
If watering had no relationship with bee visits within each habitat, we'd see an F at least this large with probability 0.013, so we reject H_0 and conclude there is evidence for bee visits varying with watering after accounting for habitat.

Part 3: Three levels plus a continuous predictor

Now the second predictor is continuous. Does the categorical predictor justify its cost *after* accounting for the continuous one? Same tasks as Part 2.

3A. Birds in city parks (together)

City parks can be refuges for birds, and bigger parks usually hold more species. Does what surrounds a park matter too? A city surveyed bird species **richness** in 8 parks in each of three surrounding land uses (**land_use**): residential, commercial, and industrial. It also recorded each park's **area** (ha).



Full model

$$\text{richness} = \beta_0 + \beta_1 \text{area} + \beta_2\,(\text{land use = Commercial}) + \beta_3\,(\text{land use = Industrial}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

Draw the models

```
coef(lm(richness ~ area, parks))
```

```
(Intercept)        area
  9.2357591   0.7618654
```

```
coef(lm(richness ~ area + land_use, parks))
```

```
      (Intercept)              area land_useCommercial land_useIndustrial
        9.9183439         0.8657746         -1.2911902         -4.2505121
```

See figure above

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`richness ~ area`	2	22	257.9
`richness ~ area + land_use`	4	20	188.4

Adding land use costs 2 df (two dummy variables).

$$F = \frac{(257.9 - 188.4)\ /\ (4-2)}{188.4\ /\ (24-4)} = 3.69$$

Interpretation

State H_0 and H_A.

H_0: $\beta_2 = \beta_3 = 0$

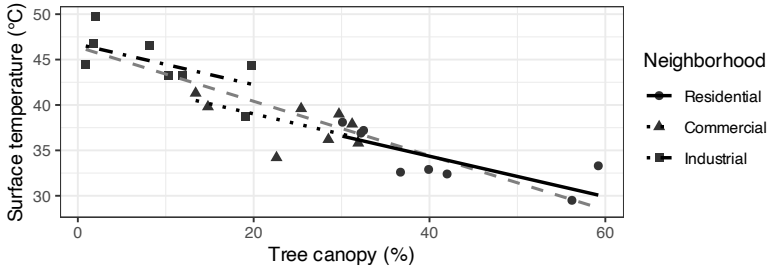
H_A: at least one of β_2 and β_3 is different than 0.

The p-value for this F is $p = 0.043$. Interpret it in a sentence.

If land use had no relationship with richness among parks of the same area, we'd see an F at least this large with probability 0.043, so we reject H_0 and conclude there is evidence for bird richness varying by land use after accounting for park area.

3B. Urban heat (in groups)

Pavement and rooftops soak up summer sun, while trees shade the ground. A city measured summer surface temperature (**temp**, °C) at 8 sites in each of three **neighborhood** types: residential, commercial, and industrial. It also recorded the tree **canopy** cover (%) around each site. Does neighborhood type matter *after* accounting for canopy?



Full model

$$\text{temp} = \beta_0 + \beta_1 \text{canopy} + \beta_2\,(\text{neighborhood = Commercial}) + \beta_3\,(\text{neighborhood = Industrial}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

Draw the models

```
coef(lm(temp ~ canopy, sites))
```

```
(Intercept)      canopy
 46.3901866  -0.2991744
```

```
coef(lm(temp ~ canopy + neighborhood, sites))
```

```
          (Intercept)                canopy neighborhoodCommercial neighborhoodIndustrial
           43.2780196            -0.2230053              0.2024248              3.4194922
```

See figure above

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`temp ~ canopy`	2	22	122.1
`temp ~ canopy + neighborhood`	4	20	96.1

Adding neighborhood costs 2 df.

$$\text{F} = \frac{(122.1 - 96.1)\ /\ (4-2)}{96.1\ /\ (24-4)} = 2.70$$

Interpretation

State H_0 and H_A.

H_0: $\beta_2 = \beta_3 = 0$

H_A: at least one of β_2 and β_3 is different than 0.

The p-value for this F is p = 0.091. Interpret it in a sentence.

If neighborhood type had no relationship with temperature among sites with the same canopy, we'd see an F at least this large with probability 0.091, so we fail to reject H_0. There is not enough evidence to

conclude that temperature varies by neighborhood type after accounting for canopy.